Do Visual-Selection Heuristics Measure Technical Quality? A Controlled Audit of Sixteen Operations Used in Video Summarization

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Abstract

Video summarization systems select segments using visual heuristics such as colourfulness, contrast and sharpness in heterogeneous roles, from features of a photograph classifier to terms inside summarization objectives. We audit their reuse as autonomous technical-quality scores. **Of the sixteen individual operations, only one is stand-alone in its own source**, and it indexes colourfulness. We reimplement fifteen literature-derived heuristics from eleven source works, plus ITU-T spatial information as a diagnostic sixteenth operation, and screen them on 800 pairs in which the same content appears before and after a known degradation, so the better member is fixed by construction. **Thirteen prefer the degraded copy under white noise**, and the Bag-of-Visual-Words score deployed in a published summarizer inverts on 92% of those pairs. An exact factorization explains the failure, and it is the count channel rather than the rarity weight that dominates: count alone predicts the sign in 797 of the 800 pairs. None of the fifteen directionally evaluated heuristics is eligible under the screening rule, and neither are the nineteen evaluated combinations; three of ten no-reference models are. Substituting an eligible model into an otherwise fixed pipeline changes 22 of 42 selected segments. For the technical-quality requirement evaluated here the three eligible models are better-supported replacement candidates, and **the quality operation should be reported as a design choice** evaluated against the requirement it serves, rather than an implementation detail.

Keywords: Multi-video summarization, No-reference image quality assessment, Content selection criteria, Benchmark evaluation, Broadcast news video

1 Introduction

Video summarization reduces long audiovisual collections to selected keyframes, shots or segments [3, 56]. When several sources cover the same topic, selection must also handle cross-source redundancy [36, 44, 62]. A multi-video pipeline therefore combines several criteria: a segment can be relevant yet redundant, visually informative yet technically poor, or temporally misplaced.

One stage of such a pipeline scores segments on their visual content, and the works that supply operations for that stage read it as a quality filter. Colourfulness, contrast, edge distribution, hue count, sharpness, entropy and saliency area recur there as selection or quality-related terms [8, 24, 29, 33, 47, 53], in heterogeneous roles: a standalone score for the frame, a term inside a larger objective, or a threshold that discards. Several originate elsewhere, in photo-quality classification [32], colourfulness measurement [21], retrieval [16, 51] and spatial-complexity characterization [26], each validated against a different question. **Treating the individual terms as autonomous, higher-is-better technical-quality scores is the reuse this article audits.**

The test needs material where the answer is not a matter of opinion. We build 800 pairs in which the same broadcast content appears before and after a deterministic degradation, so the member a quality score should rank higher is fixed by construction, and we run fifteen operations reimplemented from eleven source works, plus the ITU-T spatial information index, through it. **Every operation whose source declares a direction ranks the degraded copy higher on at least one degradation type**, beyond sampling error in fourteen of the fifteen, and thirteen of them do so under white noise. For the deployed score the cause is exact: it factors into a descriptor count times a mean weight, and the count carries the sign of the difference in 797 of the 800 pairs. For the others the article reports a pattern rather than a mechanism: most are statistics of local structure, and additive noise raises those responses.

Auditing a class needs a system where the criterion is stated rather than implied, and HSMVideoSumm [5, 6] is modular enough that the visual-content operation can be exchanged while every other stage is held fixed. Its criterion for that stage is human-derived and concerns visual relevance, not image quality; Section 2 states it. **What this article evaluates is the reading the later literature applies to that stage**, scoring the technical quality of candidate representatives, and how far a different operation moves the selection.

The scientific issue is the distinction between that criterion and the operation that implements it. The implementation is a later engineering choice: Scale-Invariant Feature Transform (SIFT) descriptors are quantized through Bag-of-Visual-Words (BoVW), weighted with term frequency–inverse document frequency (TF-IDF), and summed [16, 38, 50, 51]. SIFT, BoVW and TF-IDF were developed for matching, retrieval and categorization, not for predicting perceptual quality. Calling their aggregate a quality score does not establish construct validity [15], which concerns whether an operational measure represents the construct attributed to it.

No-reference image quality assessment (NR-IQA) provides alternatives because segments from different broadcasters have no common undistorted reference. The available families include natural-scene-statistics, supervised convolutional and transformer regressors, and vision–language models [1, 31, 39, 40, 69]. The question is not

whether a newer model produces a higher number, but whether the deployed operation responds to the construct it names, whether that behavior can be explained, and whether replacing the operation changes the pipeline output.

This study makes four contributions: a provenance-based inventory of visual-selection operations and the roles their source works define for them, each fixed by a verbatim excerpt; a controlled evaluation of degradation response and cross-database generalization, including a mechanistic analysis of the deployed BoVW+TF-IDF component; an assessment of the combination rules the sources specify, with our own adaptations identified as such; and an operation-level substitution study measuring changes in representative selection within a fixed multi-video pipeline. Training-database overlap, detectable-keypoint conditioning, alternative pooling, overlay strata, registered predictions and computational details are diagnostic controls and are reported in the supplementary material.

2 Related work

2.1 Quality-aware multi-video summarization

Extractive summarization selects existing frames or temporal segments, whereas abstractive approaches generate derived representations [3, 56]. Multi-video methods add explicit cross-source redundancy and diversity. Video-MMR formalized the setting [36]; sparse and collaborative formulations improved diversity across topic-related videos [44, 45]; graph models subsequently represented inter-video relations directly [62]. These methods address representativeness, coverage and redundancy, but they do not validate a technical-quality proxy used to choose between redundant candidates. Selecting among concurrent recordings of one event is an established setting in this journal: ViComp composes user-generated videos of the same event [4], and Cricri et al. [14] extract events and recording-activity information from that same material. **This article adds no selection rule to that line**; it tests whether the operation such a rule calls tracks the construct it is read as measuring, and what changes downstream when it is replaced.

HSMVideoSumm instead derived selection criteria from human summarization practice and implemented each criterion as an explicit pipeline stage [5, 6]. A study with human summarizers of multi-source news derived thirteen criteria, eight for inclusion and five for exclusion, and the system implements ten of them across five stages. **The source defines the criterion of the stage audited here as whether a segment *contains images of interest***: a relevance criterion, and the same study reports that aesthetic quality was not relevant to its participants. The modular design enables controlled substitution of the visual-selection operation: the same clusters, filters and ordering can be held fixed while any candidate operation is dropped into the visual-selection stage, so fifteen operations from eleven works, plus ITU-T spatial information as a diagnostic control, can be compared under conditions none of their source papers shares. The object of comparison is therefore neither a new summarizer nor an overall benchmark against SumMe [20] or TVSum [52]. It is a decision that recurs across this literature, namely which of two redundant segments is visually better, together with the operations the literature offers for making it. Otani et al.

[43] make the companion point on the evaluation side, showing that segmentation and protocol choices can dominate reported summarization scores; **the selection side carries the same exposure**, which is what an operation-level audit measures.

The works audited here place that decision in three different structures, and the structure is what the later reading flattens. **Two combine several terms into a single frame-level quality score**: Srinivas et al. [53] averages five attributes into a per-frame quality score, which is itself one term of the weighted sum that orders frames there, and Korolkov [33] weights sharpness and brightness as $0.7\,z(\text{sharp}) + 0.3\,z(\text{bright})$. **Five carry a visual term inside a larger objective**: Qaroush et al. [47] distributes four terms across visual attention, coverage and diversity; Jeong et al. [29] places a badness term inside a cost multiplicative in cluster distance, whose pixel-variance component recurs in Xu et al. [63]; Yang et al. [64] adds one texture term among several; and Huang et al. [24] one aesthetic attribute among several, as Gygli et al. [20] does with three averaged. **One uses the statistic as a gate**: Cakmak et al. [8] thresholds Laplacian variance to discard blurred frames.

Two further works supply attributes without being summarizers, and separating them matters for what any result can be attributed to. Ke et al. [32] designed edge distribution, hue count, contrast and colourfulness as features of a naive Bayes classifier that separates professional photographs from snapshots, and Hasler and Süsstrunk [21] published a colourfulness index fitted to a psychophysical experiment. Later summarization work borrows their expressions; the borrowing, not the original design, is what this article tests.

The use of image-quality assessment in summarization is not itself new. VSQUAL used objective IQA measures for shot-boundary detection, keyframe extraction and summary construction [13]. Jeong et al. also combined representativeness with contrast, blur and image-skew measures when selecting keyframes for static video summaries [29]. The present contribution is a mechanistic construct-validation audit of an already deployed quality operation, followed by dependency-aware propagation of an operation-level substitution.

2.2 No-reference quality assessment and transfer

Image quality assessment predicts perceived degradation. Full-reference measures require an undistorted original and are unsuitable when candidates come from different broadcasters [59, 67]. NR-IQA instead estimates quality from the test image alone. BRISQUE and NIQE model departures from natural-scene statistics [39, 40]; DBCNN and HyperIQA learn opinion-supervised convolutional representations [55, 68]; MUSIQ and TOPIQ use multi-scale or top-down attention [12, 31]; saliency has also been used to condition a convolutional predictor [30], and self-supervised representations have been applied to tone-mapped images [10].

Vision–language formulations based on Contrastive Language–Image Pre-training (CLIP) [48] provide a different inductive bias, and three of the candidates below share that backbone. CLIP-IQA uses antonym prompts [57]; LIQE jointly predicts quality, distortion and scene [69]; QualiCLIP learns from rankings of synthetically degraded pairs without opinion scores [1]. Q-Align maps multimodal-model outputs to discrete quality levels [61]. These models differ not only architecturally but also in their

exposure to opinion databases. A result on training data measures fit, whereas use in a new broadcast domain requires transfer. Cross-database evaluation is therefore essential, and published work has shown that in-database NR-IQA performance can overstate generalization [9, 46]. Improving the precision of objective quality meters, rather than only applying them, is a stated concern in this journal [7], as is assessing a measure through paired comparisons instead of an absolute scale, whether by learning to rank degraded variants [28] or by regressing on pairwise score differences [42]. **The constructed pairs used here are that design applied to the operations a summarizer runs.**

KADID-10k supplies controlled synthetic distortions [37]; KonIQ-10k and LIVE In the Wild (CLIVE) contain authentic in-the-wild images [18, 22]. Their differences make them complementary validation settings, but comparisons among unmatched databases and checkpoints do not identify causal effects of training exposure or distortion regime. The complete descriptive contrasts are consequently kept in Supplementary Table S15 rather than presented as a principal contribution.

Two further cautions affect the broadcast application. First, professional and user-generated video can contain temporal impairments that a frame-wise image score cannot observe; video-quality models were developed partly for this reason [25, 58]. Second, high correlation on a reused database need not imply difficult pairwise discrimination. Database range and subjective-score uncertainty affect correlation [34], and recent multimodal benchmarks report that comparative quality judgments can behave differently from absolute scoring [70]. Newer distributional quality models also exceed earlier reference-model results on some databases [66]. Accordingly, Q-Align and DOVER are treated here as complementary contemporary references, not definitive ground truth or evidence that temporal modeling itself causes an advantage.

2.3 The unresolved measurement problem

SIFT, BoVW and TF-IDF encode local-feature occurrence and rarity for retrieval [16, 38, 50, 51]. Spatial Information similarly quantifies scene complexity rather than perceptual quality [26, 49]. Such features may correlate with opinion scores in a particular collection, but correlation does not show that they measure quality or remain valid under controlled degradation. What counts as quality is in any case defined against the task the measure serves, and task-based optimisation makes that dependence explicit [35]. **The requirement audited here is therefore not the one the source states**: the source states relevance, and what this article tests is the technical-quality reading that later work applies when it carries these operations into that stage. Whether an operation serves the relevance criterion its source defines is a different question, and not one these measurements answer.

The gap addressed here is narrower than the general problem of quality-aware summarization. Existing work establishes multi-video selection criteria, specialized NR-IQA methods and prior use of IQA in summarization, yet the evidence the sources themselves offer for that reading is thin. Only two of the ten source works read in full validate the named visual property directly, and one of those two names colourfulness rather than quality; the summarization works evaluate the resulting summary rather than the visual component in isolation, and Korolkov [33] reports no isolated test of

its weighting rule. **Agreement with a reference summary is evidence about a summary, not about a quality measure**, and it is as a quality measure that these operations are used. This study puts sixteen of them through one protocol, described in Section 3, which pairs a provenance audit against the sources with four forms of measurement. That combination **distinguishes construct validation from merely integrating an available IQA model**, and it separates a defect of the operation from a defect of how the later literature aggregates it.

3 Audit design

The object of the audit is a class of operations, not a system. Fifteen visual operations that the summarization literature applies at the selection stage, plus ITU-T spatial information as a diagnostic control, are reimplemented from the equations their source works publish, fifteen of them closed-form and one requiring a saliency map, for which the source prescribes a learned backbone and this audit substitutes the spectral-residual map [23] while keeping the operation the source defines, the salient fraction of frame area; **that row is an adaptation and not a reproduction**. Every one of them is put through a common protocol (Fig. 1). Ten no-reference quality models and one deployed pipeline component are carried through that protocol, with the tests available for each implementation recorded in the coverage matrix, so that the audited operations have a contemporary quality-assessment reference.

Four forms of evidence enter. Construction-defined degradation pairs in the broadcast domain fix the better member without a reference model and give the primary test. External validation against human opinion scores on three public databases gives the severity response and its transfer. A mechanistic analysis, available in closed form for the one operation that is a published pipeline component, supplies the explanation. An operation-level substitution in a working pipeline measures how far the choice moves real decisions. A method is eligible only if it responds to controlled degradation, generalizes to every unseen opinion database and exceeds a reference-model agreement floor in the deployment domain.

Three categories of implementation are kept apart throughout. An operation is *reproduced* when the source publishes an equation this article evaluates unchanged; *adapted* when the source defines the term over a region, a sequence or a color space that the single-frame protocol requires reducing, in which case the reduction is stated; and *exploratory* when no source declares a direction, in which case the operation is reported but excluded from any pass-or-fail count. Two of the sixteen are singled out on this basis and both are stated wherever they appear: the International Telecommunication Union Telecommunication Standardization Sector (ITU-T) spatial information index declares no quality direction and is therefore excluded from every accuracy count, leaving fifteen; and saliency area is the one operation whose source requires a learned map rather than a closed-form expression, the other fifteen being closed-form. Table S23 carries the per-operation detail. Reproduced and adapted forms are counted separately and never summed.

Table S23 of the supplement carries the correspondence: for each operation, the source work, the formula as that source states it, the function that computes it, the

direction it declares and the reference model used on the in-domain column, with the results file named in its footnote. **Every operation name comes from that mapping**, and every value in this article's tables and figures is generated from the measurement files it points to. **The audited set is sixteen operations: fifteen that the summarization literature places in the visual-selection stage, drawn from eleven source works, and spatial information, which comes from the ITU-T recommendation on subjective video testing rather than from that literature.** The fifteen carry a direction their source declares and are the ones any pass-or-fail count refers to; spatial information carries none. The deployed BoVW score is *not* one of the sixteen: it is the pipeline component under audit, screened by the same protocol and reported alongside them, which is why the tables show seventeen rows where both appear.

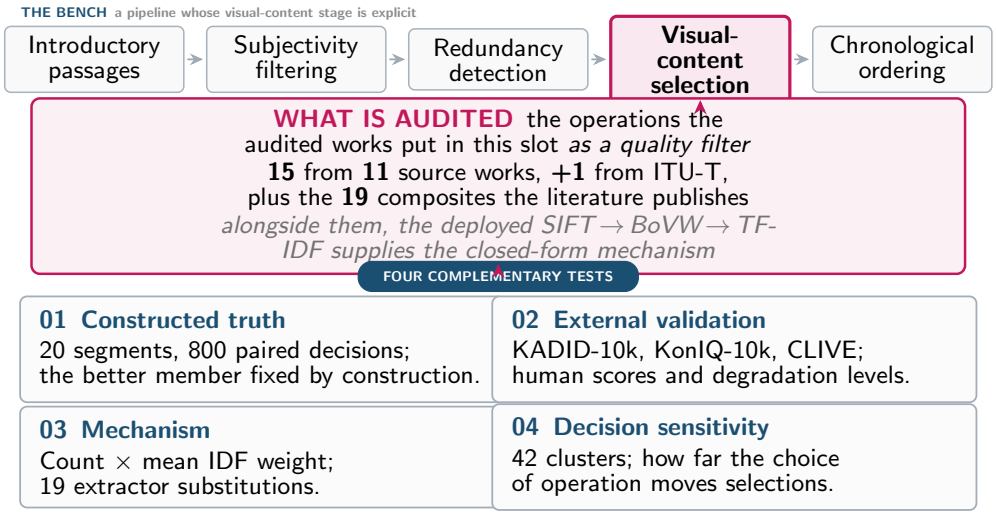


Fig. 1 Scope and evidence flow. Top: a deployed multi-video pipeline whose visual-content stage is explicit, which is what makes the slot testable. Middle: what is audited is the set of operations the summarization literature puts in that slot as a quality filter, fifteen from eleven source works plus ITU-T spatial information as a diagnostic sixteenth, and the nineteen composites the literature publishes, not the pipeline itself. Bottom: the four tests, applied to the operations, to ten no-reference models and to the deployed case, with per-test coverage recorded in Table S19 because it is not uniform. The provenance audit that produces the middle inventory is the fifth evidence block. All other pipeline stages remain fixed.

3.1 Identifying the operations, and reproducing the published rules

The operations audited here were not assembled from a database query, and the population of inference is stated accordingly. They come from a curated archive of 186 full-text PDFs on video and multi-document summarization and on image-quality assessment, searched on 5 September 2026 with thirteen patterns, one per

operation family. Each pattern is a specific expression rather than a bare word, so that *contrast* on its own never matches; reference lists are cut before matching, and one term is admitted only when its context window carries the standard that defines it. **The search instrument was itself measured against a manual reading of 21 articles**: it recovers .81 of the operations present, and .63 of what it returns is correct. The residual bias runs in favour of the instrument, because the same author performed both the reading and the inventory, so the recall figure is an upper estimate. Screening then applied one stated criterion, fixed before the measurements of Section 3.4: an operation enters only if its source declares a direction for its own statistic and scores a single frame. **15 of the 24 mapped operations meet it**, and the three that declare a direction and stay out are sequence-level rather than frame-level.

Two consequences follow, and both are reported rather than argued away. **Every count in this article is a count within that archive**, not within the summarization literature as a whole; no claim of exhaustive coverage is made, and Table S31 of the supplement states the protocol and its measured accuracy in full. And the operations screened in cover 41 of the 71 operation mentions the archive contains: of the 30 outside, 25 are not screenable by any rule at all, because the source declares no direction (14) or gives no combination rule (11). **That residue is under-specification in the sources**, and it is the same property that would stop an independent reader from reimplementing those operations. Within the same archive, 173 of the 186 works cite no dedicated no-reference quality model, and 33 of those use one of the audited operations in its place, which is a floor because a work that computes an operation without naming it is not counted. **This literature does measure technical quality**, with terms whose behaviour under degradation had not been tested.

Each operation was traced to the work the literature cites for it, and three things were recorded from that work rather than inferred: the objective it states, the role the operation has inside that objective, and a verbatim excerpt fixing both. One criterion was applied to all sixteen. **An operation is a stand-alone index in its source when it alone produces the number that orders the candidates there**; it is not when it enters as a term of a sum, a feature of a classifier, one block of a spliced representation vector, or one signal in a filter that discards frames. **The criterion applies to composites exactly as to single attributes**: the five-attribute composite of Srinivas et al. [53] is a per-frame quality score in its source, the average of its five terms, yet the number that orders candidates there is a weighted sum in which that score is one term, so the composite is not stand-alone either. Where a source names a numbered objective, the equation is cited so that each term's membership is checkable. **By that criterion one of the sixteen is a stand-alone index in its source**, the Hasler colourfulness expression, and it indexes colourfulness rather than quality; the rest enter as classifier features, terms of a sum, blocks of a spliced vector, or signals in a discard filter.

A fourth thing was recorded from every source whose full text is in our archive: the evidence the work itself reports for its visual criterion. For each we noted **the object the source evaluates**, whether an individual attribute, a combined module or the final summary; **the reference it evaluates against**, whether human ratings of the named property, summary-level evaluation, or an internal ablation; and a verbatim

excerpt fixing both. **Ten of the eleven source works were read in full for this dimension**; the exception is Xu et al. [63], one of three sources for contrast variance, whose full text is not in our archive and which is therefore **marked as not assessed rather than as reporting no validation**, since the two are different claims. Table S26 of the supplement carries the classification with its excerpts.

Where a source also states *how* its terms combine, that specification was implemented as written and put through the paired test of Section 3.4 with the terms, frames and pooling held fixed, so the rule is the only thing that differs. Source-specified rules are reproduced only when every quantity they require is available on the paired bench; otherwise the relationship is reported structurally rather than evaluated experimentally, as with a term needing a frame's distance from its cluster center.

Table S17 states, for each of the sixteen operations, the objective its source declares and the role it has there. The reading tested here is the one the summarization literature applies when it carries these operations into a selection stage: one frame, no model, higher is better technical quality.

Table S25 of the supplement links each source-defined role to the implementation evaluated here, including changes of input, aggregation and score direction, with the verbatim excerpt that fixes each role. Every one of the sixteen operations carries such an excerpt, spatial information included: ITU-T P.910 defines it as *"a measure that generally indicates the amount of spatial detail of a picture"*, higher for more spatially complex scenes, and uses it to select test sequences [26]. Table 3 reports the corresponding results by source work and component.

The four operations taken from Qaroush et al. [47] show the distinction at its sharpest: that source states four numbered objectives and its four terms do not share one. Gradient magnitude and saliency enter visual attention (Eq. 20), but entropy enters segment coverage (Eq. 21) and gradient orientation segment diversity (Eq. 22), so two of the four are not quality terms in their own source at all. Two works that adopt the same operation can also carry it differently: Laplacian variance is a discard threshold in Cakmak et al. [8] and a weighted term in Korolkov [33], and in neither is it the score that orders candidates.

Jeong et al. [29] show the same distance in another form: contrast there is pixel variance, but it enters a *badness* beside blur and skew inside $a_i = d_i\sqrt{c_i + b_i + g(c_i + b_i + s_i)}$, whose leading factor is the Mahalanobis distance of the frame from its cluster center. **The component audited here is that grey-level dispersion on its own**, read in the increasing direction and computed as the standard deviation, which is the reading common to the two sources that state it (Table S23).

3.2 Data and units of analysis

The external benchmark uses KADID-10k, KonIQ-10k and LIVE In the Wild (CLIVE). KADID-10k contains 10,125 distorted images derived from 81 references, with 25 distortion types at five levels and Differential Mean Opinion Scores (DMOS) [37]. KonIQ-10k contains 10,073 authentically distorted images, approximately 1.2 million ratings and Mean Opinion Scores (MOS) [22]; CLIVE contains 1,162 mobile photographs with authentic distortions [18]. No no-reference model is

retrained or fine-tuned. Two *combination* rules are fitted over the operations' outputs alone: the ridge variant of Section 4.3 and the classifier reproducing Ke's rule, validated leaving one segment out.

The deployment analysis uses the redundancy stage produced by HSMVideoSumm over 19 of its 21 news topics: 42 multi-member clusters, 93 segment slots and 83 distinct segments. The other two topics contain no multi-member cluster. These clusters are operational inputs, not proof of equal semantic utility. The original human redundancy annotations cover only 15 clusters and are retained as a supplementary sensitivity analysis (Tables S1, S7, S8 and S11); the 42-cluster set is the primary analysis because it is the set encountered by the deployed pipeline. Throughout, a *sensitivity analysis* repeats a reported result under an alternative choice, and a *diagnostic control* probes a mechanism without carrying a pass-or-fail verdict.

Still-image scorers receive frames sampled uniformly at 1 frame per second and are pooled by the arithmetic mean,

$$S(s) = \frac{1}{n_s} \sum_{i=1}^{n_s} \phi(f_i), \tag{1}$$

where ϕ is the per-frame score. The deployed operation instead applies its own keyframe selector, quantizes retained descriptors into a vocabulary of $K = 300$ visual words and sums

$$S_{\mathrm{BoVW}}(s) = \sum_{i} \sum_{v=1}^{K} \mathrm{tf}(v, f_i)\, \mathrm{idf}(v). \tag{2}$$

Sampling, scoring and pooling jointly define the operation occupying the visual-selection stage, so the primary comparison evaluates complete operations while holding every other stage fixed. A matched-input control additionally gives SIFT+BoVW and LIQE the same fixed set of at most ten uniformly spaced 1-fps frames and uses arithmetic-mean pooling for both. Full scores and alternative normalizations are in Tables S6, S9 and S11.

3.3 Scorers, references and polarity

The complete-case comparison contains the deployed SIFT+BoVW score and ten NR-IQA candidates: BRISQUE, NIQE, DBCNN, HyperIQA, MUSIQ, TOPIQ, CLIP-IQA, CLIP-IQA+, LIQE and QualiCLIP. Keypoint density, Spatial Information, deep feature norms and the second IDF formulation are diagnostic controls. A separate ablation replaces SIFT with 19 convolutional neural network (CNN) extractors while holding the vocabulary, weighting and aggregation fixed. ARNIQA and TReS extend coverage on completed axes but are excluded from complete-case screening; MANIQA was not completed before the experimental freeze (Table S13). Released-weight supervision, checkpoint role, polarity and graphics processing unit (GPU) requirements are summarized in Table 1 and listed in full in Table S13 [2, 19, 65].

Q-Align is an in-domain reference model rather than a candidate [61]. **The two families are scored against different references on the in-domain axis**: the ten no-reference models are compared with Q-Align, and the summarization operations

Table 1 Scorers needed to interpret the main comparison. Sup.: KQ = released KonIQ-10k opinion-trained checkpoint; LV = LIVE opinion training; OU = opinion- unaware; Mix = multi-database opinion mixture; – = no opinion data. **KQ is the supervision code, distinct from the vocabulary size K of Section 4.2.** Polarity is declared before evaluation. D = deployed, C = candidate, R = reference, X = diagnostic or ablation. Partial-coverage methods remain outside screening.

Method	Family	Measured construct	Sup.	Pol.	GPU	Role
SIFT+BoVW	handcrafted retrieval	TF-IDF-weighted visual-word count	–	↑	no	D
Keypoint density	handcrafted	detectable keypoints per frame	–	↑	no	X
DINOv2 [41]	self-supervised features	mean norm of patch tokens	–	↑	yes	X
Spatial Information	ITU-T complexity	Sobel energy	–	–	no	X
BRISQUE	natural-scene statistics	deviation from learned natural statistics	LV	↓	no	C
NIQE	natural-scene statistics	distance from a pristine model	OU	↓	no	C
DBCNN	CNN regression	bilinear distortion/content features	KQ	↑	yes	C
HyperIQA	CNN regression	content-adaptive quality regression	KQ	↑	yes	C
MUSIQ	transformer	multi-scale image quality	KQ	↑	yes	C
TOPIQ	transformer	top-down quality attention	KQ	↑	yes	C
CLIP-IQA	vision–language	antonym-prompt quality	OU	↑	yes	C
CLIP-IQA+	vision–language	learned-prompt quality	KQ	↑	yes	C
QualiCLIP	vision–language	degradation-ranked quality	OU	↑	yes	C
LIQE	vision–language	quality, distortion and scene	KQ	↑	yes	C
Q-Align	multimodal language model	discrete textual quality levels	Mix	↑	yes	R
19 CNN+BoVW	descriptor ablation	TF-IDF-weighted visual-word count	–	↑	yes	X
ARNIQA; TReS	partial coverage	distortion manifold; ranking consistency	KQ	↑	yes	C

The 19-extractor ablation comprises ResNet-18/50/101, VGG-16/19, EfficientNet-B0/B4, MobileNet-V3-Large/V2, ConvNeXt-Tiny/Small, DenseNet-121/201, ResNeXt-50/101, Wide-ResNet-50/101, Inception-V3 and SqueezeNet-1.1. Released weights, endpoints and preprocessing are in Table S13.

with LIQE, so that column compares each family with a reference outside its own kind and never across the two. The `q-future/one-align` checkpoint is run with the `quality` task, `image` input, 4-bit NormalFloat4 (NF4) double quantization and 16-bit floating-point (FP16) compute, then pooled by Eq. 1. It was among the strongest public NR-IQA models when the inventory was frozen, is architecturally distinct from the candidates, and is 13.7 times slower than LIQE under the measured setup; it is therefore reserved as a reference model rather than screened as an on-line candidate.

The substitution study measures changes of selection and model-reference scores of the output; **it does not measure human preference** for the resulting summaries. DOVER is used only as an output-sensitivity reference and does not participate in screening [60]. The official `DOVER.pth` checkpoint and `dover.yml val-l1080p` views are used (Table S13). Its raw technical branch is primary; the uncalibrated technical–aesthetic sum is only a sensitivity analysis, and neither the official standardized fusion nor DOVER++ is computed. The same technical output is tested on the 800 constructed pairs, using a common 5-fps MP4 wrapper and paired fragment seed 42. Because DOVER was developed for user-generated video, its output is not treated as broadcast-domain human ground truth.

Score and target directions are declared before analysis. Direction-aligned correlation is $\rho^* = s_m s_y \rho$, where $s_m = +1$ for a higher-is-better method and -1 for a lower-is-better method, while $s_y = +1$ when a higher target means better quality and -1 when it means worse quality. All opinion targets used here, including KADID's released `dmos` field, are higher-is-better; degradation level is higher-is-worse. A negative ρ^* therefore denotes a genuine directional inversion rather than a scale convention.

3.4 Controlled degradation and constructed truth

KADID sensitivity is computed without treating derivatives of the same source as independent. For each distortion type, Spearman correlation between score and level is computed within each of the 81 references and averaged across references. Confidence intervals use 10^4 block-bootstrap resamples of the reference images with seed 42.

The broadcast experiment supplies a second controlled test. Eight degradations are applied at five levels: Gaussian blur, white noise, darkening, JPEG compression, low-bitrate H.264 transcoding, a downscale-and-restore round trip at five factors (Table S2), horizontal motion blur and progressive desaturation. The last uses $S'_\ell = \mathrm{clip}(\alpha_\ell S, 0, 255)$ with $\alpha_\ell \in \{0.75, 0.55, 0.35, 0.18, 0.05\}$. H.264 is applied to the frame sequence; other transformations are frame-based.

Twenty of the 83 broadcast segments are sampled without replacement by a seeded simple random draw (seed 42); they represent 13 of the 19 operational topics (Table S2). Pairing each unmodified segment with every degradation and level gives $20 \times 8 \times 5 = 800$ binary decisions. The unmodified member is the construction-defined higher-fidelity signal. Observations are nested, and the unit is therefore reported per analysis rather than once for the design: strict accuracy is block-bootstrapped by topic and the paired score difference by segment (10^4 resamples, seed 42), because the 40 pairs inside a segment share content and the 20 segments come from 13 topics. **Per-cell significance uses an exact sign-flip randomization over the 20 segments**, whose null enumerates all 2^{20} block sign assignments, with Holm over the 120

operation–type tests. **Disagreement counts between the complete-operation and matched-input regimes are compared by an exact McNemar test** over the 42 paired clusters and, because those clusters come from 19 topics and McNemar treats them as independent, by the same block sign-flip over topics, which enumerates all 512 assignments the discordant pairs admit across 9 topics. **Primary inference uses the dependence-aware unit each design implies**: the reference image on KADID-10k, the topic on constructed-pair accuracy, the segment on the paired difference, and the news topic on the application. **Each unit follows from the design rather than from agreement with a published bound**: observations that share content or a source enter the same block, which is the segment for pairs cut from one segment and the topic for clusters drawn from one news topic. Table S27 additionally recomputes every interval under the rival unit; **that comparison is a verification, not a reason for the choice**, and it identifies which implementation produced a published bound. A second design correlates score with degradation level within each segment–type pair. Overlay strata, the complete registered prediction audit and parameters are reported in Tables S2 and S12 and the reproducibility archive. Ties count as neither correct nor incorrect and remain in the denominator. A sensitivity analysis excludes darkening and desaturation, whose transformations define signal change more directly than universal perceptual worsening. These pairs test fidelity response, not preference between distinct broadcasts.

3.5 Screening and application inference

The screening rule has three conditions. Sensitivity passes when the within-reference mean is at least .50 and at least 15 of 25 distortion types exceed .50. Generalization passes when the direction-aligned Spearman Rank Correlation Coefficient (SRCC) is at least .65 on *every* database absent from the released checkpoint's opinion training. The in-domain floor is confirmed when normalized agreement with the reference designated for that family above, Q-Align for the no-reference models and LIQE for the operations, is above chance and the lower bound of its 95% topic-block bootstrap interval exceeds zero on the common 42 clusters. Cost is reported but not thresholded. Table S10 varies the numerical thresholds and the quantifier over unseen databases.

For cluster c with k_c candidates, independent uniform choices agree with probability $1/k_c$. The common-set floor is therefore $\bar{p}_e = |C|^{-1} \sum_{c \in C} 1/k_c = .470$. Normalized agreement is $\kappa_{\text{uniform}} = (p_o - \bar{p}_e)/(1 - \bar{p}_e)$. This is a custom uniform-choice correction, not Cohen's kappa. A positive point estimate is reported separately from a confirmed pass so that a wide interval crossing chance cannot be read as affirmative evidence.

External association uses Spearman correlation against MOS or DMOS. Confidence intervals (CIs) for KonIQ and CLIVE use the Fisher transform; KADID intervals and paired differences use reference-block bootstrap. Dependent correlations on the same images use Steiger's statistic [54]. The operational equivalence margin is $|\Delta\rho^*| < .02$, chosen as the largest association difference treated as practically negligible for this diagnostic rather than as a literature-wide standard; the paired difference and interval are also reported so stricter margins can be applied. The multiplicity family contains all 630 pairwise contrasts among 36 evaluated scores (17 methods and 19 ablations); Bonferroni correction gives $\alpha = 0.05/630$ and critical $|z| = 3.95$. The full accuracy,

consistency and linearity triad after the monotonic mapping required by ITU-T P.1401 is reported in Table S4 [27].

For each operation, the application selects one representative per cluster. If y_c, l_c and h_c are the selected, lowest and highest reference scores in cluster c, define $\bar{y} = |C|^{-1} \sum_{c \in C} y_c$ and analogously $\bar{l}$ and $\bar{h}$. The reported global-achievable-range estimand is

$$100 \, \frac{\bar{y} - \bar{l}}{\bar{h} - \bar{l}}. \tag{3}$$

It is a ratio of means, not an average of independently rescaled clusters. Raw branch-score differences, mean per-cluster position, selected-maximum rate, range distributions and exclusions of the smallest 10% and 25% of ranges are reported in Table S11. Zero-range clusters are undefined for per-cluster position and are counted explicitly. The primary interval resamples the 19 topics and retains their clusters. A hierarchical sensitivity analysis additionally resamples clusters within each sampled topic. Both use 10^4 paired resamples and seed 42. Human compositional preference was not collected; the estimand is explicitly a model-reference output.

Experiments ran on one NVIDIA RTX 4060 GPU (8 GB). NR-IQA implementations use the PyIQA toolbox [11]; SIFT uses OpenCV and the vocabulary uses mini-batch K-means with $K = 300$ and `random_state=42`. Q-Align is quantized to 4-bit NormalFloat4 with double quantization and FP16 compute to fit the GPU [17]; the official PyIQA architecture supports fp16, 4-bit and 8-bit inference, with NF4 used in 4-bit mode. No numerical equivalence to fp16 is assumed. Frames are decoded once and cached, and latency is re-measured in one software environment after ten warm-up and sixty timed iterations. The archive includes the scoring scripts, derived results, cluster definitions, annotations, seeds and environment information; redistribution of source news video remains subject to the broadcasters' terms. The final method inventory and analysis archive were frozen on 23 August 2026. **Two dates bound different stages**: the operation inventory, the scoring runs and the analysis archive were frozen on that date, and the archive search of Section 3.1 was performed afterwards, on 5 September 2026, to document how the already-frozen inventory relates to what the archive contains. **It added no operation and changed no measurement**; its counts describe coverage. The archive reconstructs the reported statistics from hashed score caches and exposes the inference settings.

4 Results

The evidence comes in five blocks: what the sources state (Section 3.1), the operations on constructed truth, alone (Section 4.1) and as published composites (Section 4.3), the mechanism of the deployed case (Section 4.2), the class against public databases (Section 4.4), and the effect on the pipeline's own selections (Section 4.5).

4.1 Responses to controlled degradation

Eight hundred pairs make the answer independent of any reference model: each pair holds the same broadcast content before and after a deterministic degradation, so the member a quality score should rank higher is fixed by construction. Figure S4 of

the supplement gives every operation on every degradation type as a single heatmap ordered by worst cell, and Table 3 in Section 4.6 regroups the same accuracies by the work that uses each one.

Every one of the fifteen operations with a declared direction ranks the degraded copy above the original on at least one degradation type, and thirteen do so under white noise, where five of them are wrong on all one hundred pairs: gray-level co-occurrence matrix (GLCM) contrast, entropy, Laplacian variance, gradient orientation and gradient magnitude. The two exceptions to the noise pattern, hue count and edge distribution, fail elsewhere: hue count on five of the eight types and edge distribution on H.264 and JPEG. Each operation–type cell is tested on the segment, the unit the design implies, by an exact sign-flip randomization over the 20 segments, with Holm over the 120-test family: 22 of the 35 cells below chance survive it, in fourteen of the fifteen operations. The same test leaves seven of the ten published models with no type below chance. Table S28 lists the 35 cells, and removing the two degradation types aligned with an operation's own attribute leaves all fifteen still inverting.

White noise concentrates the failures. These operations are statistics of local structure such as gradient magnitude, edge density, entropy, contrast and descriptor counts, and additive noise raises such statistics, so an operation reading more structure as more quality prefers the noisier frame. Section 4.2 measures one instance of that reasoning for the deployed BoVW score, which is not one of the fifteen: there the statistic is a descriptor count, and the sign of the score difference follows the sign of the count difference in 797 of the 800 pairs, and the three exceptions differ by at most four descriptors against a median count of 1,092. **Across the fifteen the outcome recurs without that demonstration.**

On material where the better member is known, the reading tested here inverts.

The same 800 pairs place the deployed operation and the no-reference models (Table 2), with the reference model out of the decision: each pair contains the same content before and after a known degradation. The deployed operation is correct on 65.1% of 800 pairs. LIQE reaches 99.5%; the paired difference is +34.4 percentage points, segment-block 95% CI $[+29.8, +38.9]$. Every evaluated NR-IQA candidate exceeds the deployed score. **The deployed operation is not an outlier in the class it comes from**: eight of the seventeen literature-derived scorers and composites with complete paired coverage score below it, and their median is 66.0.

No method ties on these pairs. Excluding darkening and desaturation leaves 600 comparisons and does not repair the deployed operation: its accuracy is 63.7%, whereas LIQE reaches 99.8% and Q-Align 100% (Table S12).

The average conceals the diagnostic result. SIFT+BoVW is at or above 88% on five types but selects the degraded copy in 92% of white-noise pairs and 94% of JPEG pairs. Within segment, its score increases with severity for noise ($\rho = +.720$, CI $[+.455, +.925]$) and JPEG ($+.640$, CI $[+.395, +.850]$). The mechanism is directly visible in the saved descriptor counts. The degraded copy has more descriptors in 92% of noise pairs (mean change +469) and 95% of JPEG pairs (mean change +289), with mean noise count increasing at every severity level. These measurements are consistent with noise and blocking creating detectable extrema that increase the count channel in

Table 2 Strict accuracy (%) on 800 unmodified–degraded broadcast pairs: 20 segments, eight types and five levels. Values below the 50% chance floor are systematic inversions, and are set in bold as well as in colour, so no reading of this table depends on colour alone. Ties remain in the denominator.

Method	dark	motion	blur	H.264	rescale	satur.	noise	JPEG	mean
SIFT+BoVW (deployed)	100	100	90	90	88	**39**	**8**	**6**	**65.1**
LIQE	97	100	100	100	100	100	99	100	**99.5**
TOPIQ	93	100	100	100	100	93	100	100	98.2
MUSIQ	93	100	100	100	100	86	100	100	97.4
QualiCLIP	99	100	95	100	96	76	96	71	91.6
ARNIQA	91	100	99	98	100	61	82	98	91.1
DBCNN	84	97	92	97	95	92	85	81	90.4
NIQE	100	100	98	97	98	**0**	100	96	86.1
CLIP-IQA+	76	99	79	100	91	87	84	64	85.0
HyperIQA	**47**	100	100	100	100	**30**	100	100	84.6
CLIP-IQA	58	100	83	100	92	91	55	76	81.9
BRISQUE	100	100	100	99	100	**0**	67	80	80.8

Eq. 5; structure-removing distortions instead happen to move count in the direction of fidelity. The score is therefore responsive, but not consistently to quality (Table S12).

Q-Align is correct on 98.2% of these pairs (95% CI $[97.2, 99.1]$), which supports its technical-degradation competence on this broadcast content. The exact downstream DOVER technical branch (DOVER-T) output reaches 97.6% ($[96.1, 99.0]$), or 99.8% ($[99.5, 100]$) without darkening and desaturation. The 19 extractor substitutions average 44.9% (CI $[41.0, 48.9]$) and remain dominated by their descriptor count, corroborating the mechanism while refuting the registered prediction that fixed feature maps would force chance performance. The full prediction and overlay audits are in Tables S12 and S29.

4.2 Mechanistic diagnosis

The deployed inverse document frequency uses the vocabulary size, $\mathrm{idf}_{\mathrm{dep}}(v) = \log_{10}(K/\mathrm{df}(v))$, whereas canonical TF-IDF uses the number of documents, $\mathrm{idf}_{\mathrm{can}}(v) = \log_{10}(N/\mathrm{df}(v))$. Their relation is exact:

$$S_{\mathrm{dep}}(s) = S_{\mathrm{can}}(s) + D_s \log_{10}(K/N), \tag{4}$$

where D_s is the number of descriptors. The formulations therefore differ by an additive count term, not by rescaling. The identity holds within 3×10^{-11} on all 93 segment slots. At cluster scale $N \in \{2, 3, 4\}$, so the additional term rewards count and contributes a median 98.3% of the deployed score. On a database of approximately 10^4 documents, the term is negative and reverses the count ordering.

The more general factorization separates the two available channels, and Figure S1 of the supplement shows the two together with the evidence that fixes their relative

weight:

$$\text{score}_i = \sum_v \text{tf}_{iv}\text{idf}_v = D_i \underbrace{\sum_v p_i(v)\text{idf}_v}_{\overline{\text{idf}}_i}. \tag{5}$$

Thus a score can vary through how many descriptors the extractor finds and through their mean IDF weight. The relative contribution of the two channels is measured empirically.

On KonIQ-10k, the coefficient of variation is .938 for D_i and .098 for mean IDF weight. The canonical score consequently correlates .998 with descriptor count. Its association with human scores is statistically equivalent, within the operational $\pm.02$ margin, to that of keypoint density: $\rho^* = .447$ for both, with score-to-keypoint correlation .998. Steiger's $Z = .199$ is below the Bonferroni-corrected critical value of 3.95, and both one-sided equivalence tests reject. KADID gives the same association-level result under reference-block resampling: .125 for canonical BoVW and .123 for keypoint density, paired difference $+.0024$, 95% CI $[-.0019, +.0084]$. This is equivalence of measured MOS/DMOS association, not identity of the scores or constructs. The deployed formulation is almost exactly the sign-reversed count on the large databases ($-.9999$ on KonIQ, $-.9998$ on KADID and $-.9947$ on CLIVE). In the deployment regime its own descriptor count remains dominant, $\rho(S_{\text{dep}}, D) = +.994$.

Replacing SIFT with 19 convolutional extractors leaves the external ceiling where it was. Their mean $|\rho|$ is .090 on KonIQ and .112 on CLIVE, with no value above .238 on either database; KADID ranges from .156 to .418. Fixed spatial feature maps remove per-image count variation on the image databases, leaving the rarity channel measured by the ablation. On broadcast segments, however, the upstream key-frame selector varies the number of retained frames, so it reintroduces count variation: across the 19 variants, $\rho(S, D) = +.981$ to $+.997$. Exchanging the extractor therefore changes individual choices without making the ordering a better predictor of quality.

Simple normalization does not repair the mechanism. Dividing by descriptor count exposes the weak mean-IDF-weight channel already measured by the ablation. Dividing by frame count is also mismatched to the deployed operation because its internal key-frame selector makes descriptor count nearly independent of duration. On the 42 clusters, frame normalization changes the uncalibrated DOVER two-branch-sum output only from 47.4% to 50.6% and induces an inverse duration association ($\rho = -.52$). Full normalization and pooling results are in Tables S6 and S9.

4.3 Performance of literature-derived composites and source-specified rules

Nineteen attribute combinations were collected from the summarization literature (Table S30), twelve of them with more than one term. Each went through the screening rule of Section 3.5 under two aggregations chosen here, a z-score mean and, as a stronger variant, a ridge model fitted on the whole of KADID-10k; the rules the sources themselves specify are reproduced separately, later in this section.

One of the nineteen clears the sensitivity gate. The composite of colourfulness, Laplacian variance and local binary pattern (LBP) uniformity reaches a

within-reference mean of .524 on 17 of the 25 distortion types, above the .50 and 15-type requirement that no individual operation met. Its transfer then collapses: on KonIQ-10k the same composite reaches $\rho^* = .061$, and fitting the ridge on the entirety of KADID before applying it to KonIQ moves that to .067, against a requirement of .65. The drop is what the second gate exists to detect in a combination calibrated on one severity ladder. The remaining 18 do not reach even the first gate. **No combination evaluated here satisfies the protocol's two damage requirements at once.**

The two aggregations above are our choices. Several sources do specify how their terms combine, and those specifications were implemented and put through the same bench; two of them prescribe a rule over the same attribute vector as implemented here, whose adaptations are identified in Table S25 of the supplement, so the comparison isolates the rule. Ke [32] rejects the linear option in the text, writing that "a naive method might use a weighted linear combination of the quality metrics" but that "the quality metrics we defined are not linear", and uses a naive Bayes classifier instead. What is reproduced here is that rule and not Ke's fitted model: the classifier is Gaussian naive Bayes over that same vector, but its labels are this bench's original and degraded members rather than the professional and snapshot photographs Ke trained on, and its probabilities are estimated here. Validated leaving one segment out it reaches 69.9 per cent against the 87.0 per cent of the unweighted mean, a fall of 17.1 points, so the loss is attributable to the combination rule and not to a change of terms.

Srinivas [53] takes edge distribution, hue count and contrast from Ke by citation and colourfulness from a different source, and specifies the other of the two alternatives: his quality score is "the average of all these individual scores", so **the mean the later literature executes is not a simplification of that source but the rule it gives**. His weights "directly proportional to the standard deviation" belong to another level of the same paper, combining four score types of which this bench carries one. Applying that weighting to the five attributes is an adaptation of ours, not a reproduction; as a sensitivity control it reaches 84.0 per cent and gives edge distribution a weight of .00054 against the mean's .20.

One source specifies a complete module, whose **combination rule is reproduced as published over the vector implemented here**; one of its inputs, the saliency map, is the adaptation of Section 3, so the verdict below is about that implementation of the rule and not about the module in every specification its source gives. The Qaroush [47] visual-attention objective is the unweighted sum of five named terms (Eq. 20); three were already in the inventory, and the other two, the contrast term of Eq. 8 and mean motion intensity, were implemented and measured on the same bench. Over the seven degradation types that are not aligned with any member by construction, desaturation being excluded because it acts directly on the colourfulness term, the module scores 74.3 per cent while its best single term alone scores 83.1. It keeps the failure signature of the isolated operations, answering 0 per cent of white-noise and 20 per cent of JPEG pairs, and its contrast term contributes 19.9 per cent alone, which the next paragraph explains.

A second complete module behaves the same way. Korolkov [33] specifies $0.7\,z(\text{sharpness}) + 0.3\,z(\text{brightness})$ with both weights and the z-score normalization

given in the text. Normalization runs over all candidates of the bench, the original and every degraded member of each segment. On the 11 of 20 segments where both terms were measured, over the seven types that exclude darkening, which acts directly on the brightness term, it scores 64.2 per cent where its own sharpness term alone scores 69.6, and the paired contrast over the same decisions is 5.5 points with a segment-block interval of $[0.0, 10.6]$ that touches zero, so at this sample size the shortfall is not established. **For the Qaroush module the same contrast is 8.9 points with an interval of $[7.3, 10.3]$ that excludes zero: there the published module is measurably worse than its own best component.** The subset departs from the 20-segment result by 16.9 points on one degradation type, so it replicates the direction rather than estimating the size again.

In two sources the published expression carries the opposite sign to the property the surrounding text names for it. The contrast term of the Qaroush visual-attention objective is anti-correlated by construction with contrast, at -1.0000 over a sweep in which contrast rises by construction; the brightness term of the Korolkov module is stated to penalize over-exposed frames, but the mean luminance it specifies rises monotonically with exposure and enters the module with a positive weight, so brighter always scores higher. **In both cases the expression contradicts the property the same paragraph claims for it**, and in both the contradiction is visible in the equation before any measurement (Table S22 of the supplement).

4.4 External validation and screening

The constructed pairs fix the better member by construction, but they use eight degradations we chose. A public database with human opinion scores asks the same question independently, and applies to the individual operations, their composites and the no-reference models alike. All sixteen operations go through the rule of Section 3.5: two damage requirements and one in-domain requirement. Table S24 reports each family's best value on each axis, so a family fails an axis only when its best member does.

No operation clears either damage gate, and the margin is not narrow. The best within-reference mean among the fifteen is .298 against a requirement of .50, and the best lowest-unseen correlation is .139 against .65; the higher .218 in that column belongs to spatial information, which carries no verdict. Across 10,000 resamples of the reference images, none of the fifteen changes verdict and none of the intervals reaches the threshold. Four operations are negative on the first requirement, which means their response runs against the direction their own use assumes. Five of the nine operations in Table S17 that declare a direction and have a downstream choice defined do clear the in-domain requirement, and the bracketed lower bounds show how much of each pass survives its interval: the weakest is $+0.011$. Several operations therefore agree with the reference model on unmodified material while failing both external damage requirements. Scored against the same reference, all ten models clear it, at a median agreement of .708 against .281 for the operations, and two of the passing operations lose the verdict when a single topic is removed.

Within-reference KADID sensitivity separates the deployed operation from specialized quality methods. Its mean direction-aligned correlation is $-.077$ (95% CI

$[-.124, -.030]$), with only 5 of 25 types above .50. LIQE, QualiCLIP and CLIP-IQA+ reach .807, .849 and .808, respectively, each with 22 types above .50. Most remaining NR-IQA candidates also respond broadly, with means from .607 to .733.

Table S21 applies the same complete-case rule to every method. Released-weight exposure is visible because an in-database correlation is not a generalization result. The generalization column reports the lowest signed ρ^* over every unseen database, rather than using an absolute value that could hide inversion.

Three methods are eligible: LIQE, QualiCLIP and CLIP-IQA+. QualiCLIP has the largest sensitivity mean and LIQE the largest generalization margin, so the three form an eligibility set rather than a ranking. The set is unchanged when the generalization threshold varies from .60 to .65 or the sensitivity coverage cut varies from 12 to 22 types (Table S10); at .55 MUSIQ is added, and at .70 only LIQE remains.

The database-level values show why the universal unseen-database clause matters. The three eligible methods hold above $\rho^* = .65$ on both databases, while TOPIQ, MUSIQ, DBCNN and HyperIQA are strong on the unseen authentic database and fall to between .462 and .556 on KADID; Table S21 gives every value. The three eligible methods pass the floor with overlapping topic-block intervals. LIQE agrees with Q-Align on 32 of 42 clusters ($\kappa_{\text{uniform}} = .551$, 95% CI $[.284, .786]$); QualiCLIP agrees on 28 (.371, $[.143, .597]$); and CLIP-IQA+ on 25 (.236, $[.034, .435]$). The deployed operation agrees on 21, only 1.3 decisions above the uniform expectation, and its normalized interval crosses chance (.056, $[-.308, .412]$). Results on the 15 human-annotated redundancy clusters are in Tables S7–S8, kept apart from the operational 42-cluster set.

The verdict travels with the benchmark that calibrates it. The same sensitivity requirement, computed on the constructed broadcast pairs instead of KADID-10k, admits six of the fifteen directionally evaluated operations, and none meets the requirement on both benchmarks. Spatial information clears it there too, and is reported separately as a diagnostic control that receives no screening verdict. The two measure damage response by different routes, 25 synthetic types on still photographs against eight types in the deployment domain, and the rule was applied at the same fraction of types on each.

The full P.1401 triad and database-level correlations are in Tables S4 and S15. The latter reports training-database contrasts, and conditioning on detectable keypoint density strongly attenuates the BoVW score (Table S16). Cost is also secondary to eligibility: in the common environment the deployed operation takes 182 s per cluster and LIQE 1.9 s (Table S14).

4.5 What the choice of operation does to the pipeline

This section needs no reference at all: with every pipeline stage fixed and only the visual-selection operation exchanged, **do the operations pick the same segments?**

They do not. Fourteen operations for which a downstream choice is defined produce eleven distinct summaries over the same 42 redundancy clusters. Only two groups coincide, and neither by agreement: the three brightness variants compute the same quantity under different names, and the two colourfulness implementations are the same Hasler expression. That second identity is exact, and so is one more inside the

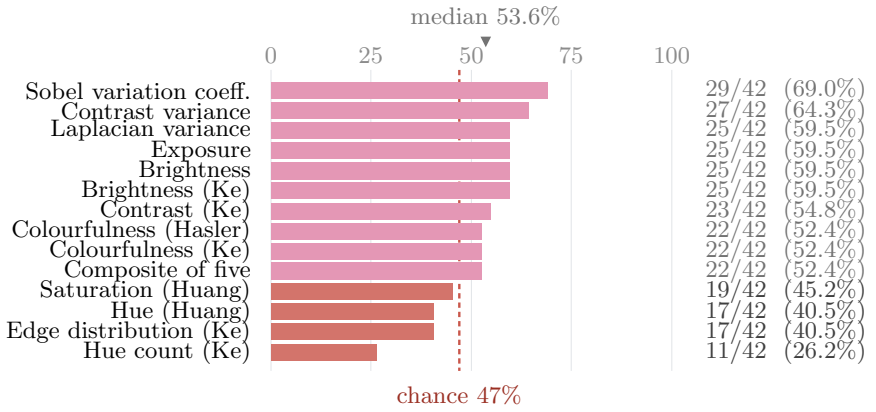


Fig. 2 What each operation does to the deployed summary. Every pipeline stage is held fixed and only the visual-selection operation is replaced; the bar is the percentage of the 42 redundancy-cluster representatives that survive the swap. **The median operation keeps 53.6 per cent of the decisions against a chance floor of 47**, and the four in red keep fewer than chance. Swapping one published operation for another therefore rewrites between 31 and 74 per cent of what the stage outputs. Both the frames and the clusters are shared across all fourteen operations and the deployed one, so nothing here comes from a difference in sampling.

audited inventory, where a further operation is the mean of three others: counted over the fifteen audited operations rather than over the fourteen exchanged here, **the raw score matrix has rank thirteen, with an effective 8 independent directions among the operations**. **That is a statement about dependence and not about the tested family**: the Holm correction of Section 3.4 is applied to the 120 operation–type comparisons actually made, and is not relaxed to 8. **In every one of the 42 clusters the fourteen select all of its members**, including the six clusters that hold three or four, so the mean of 2.21 distinct choices per cluster is the ceiling the cluster sizes allow.

Pairwise, the median agreement between two operations is 50.0 per cent against a chance floor of 47.0, and **37 of the 91 pairs agree below chance**; Figure S3 of the supplement shows both distributions. **In the same clusters, an operation agrees with the reference model about 5 times more strongly than with another operation** (median κ .281 against .056). Collapsing those duplicate computations leaves eleven distinct ones and 55 pairs, the same median of 50.0 and 23 below chance. **What this establishes is that the choice of operation changes which representatives the pipeline selects, with every other stage fixed.** No source work evaluates its operation against the others on common material.

How far the decisions move is the next question, and it applies first to the alternatives *the literature itself already offers*: a practitioner reading this literature is choosing among them, not among no-reference models. Figure 2 answers it by replacing the deployed operation with each of the fourteen in turn, holding every other stage and the decoded frames fixed.

Of the fourteen, four (Ke's hue count, Huang's hue, edge distribution, saturation) retain less than chance, so adopting one published operation in place of another can

rewrite most of what the stage outputs. Of the fourteen, nine belong to the main inventory and all nine fail the screening of Section 4.4; the remaining five are evaluated only in this substitution and carry no screening verdict. The last two columns of Table 3 link those nine to their other measurements, and Table S19 records the complete coverage. Whichever of these operations a system adopts, it inherits a different summary, and the published record offers no measurement that would let an implementer prefer one. Because these percentages use one frame per second while Table S20 scores each complete operation under its own sampling, the two must not be read across.

LIQE fills the same slot for the direct substitution: among the three eligible methods it has the largest generalization margin and near-ceiling accuracy on the constructed pairs. With the same ten uniformly spaced frames and mean pooling, SIFT+BoVW and LIQE still differ in 16 of 42 clusters. The LIQE–SIFT DOVER-T contrast is +27.1 range points (topic CI [5.6, 47.8]; hierarchical [2.4, 52.5]) and +.0074 raw, with both intervals positive. Table S20 reports the complete-operation comparison and all eligible methods.

Replacing the deployed visual-selection operation with the LIQE-based one changes 22 of 42 representatives. **The twenty unchanged clusters do not exceed chance agreement**: observed agreement is 47.6 per cent against a 47.0 per cent uniform-choice expectation, giving $\kappa_{\text{uniform}} = .011$; the topic-block 95% interval for raw agreement is [31.8, 61.7]. On the primary DOVER technical branch, the global-achievable-range point estimate rises from 50.5% to 76.0%, a paired increase of +25.5 points. Its topic-block interval is $[-12.1, +54.9]$ and its hierarchical interval is $[-17.2, +59.6]$; neither confirms a population-level effect. The raw branch-score difference is +.0069 (topic-block $[-.0028, +.0178]$; hierarchical $[-.0041, +.0197]$), while mean per-cluster position and selected-maximum rate increase by 7.2 and 4.8 points, respectively.

No cluster has zero DOVER-T range. Excluding the smallest 10% or 25% of ranges gives global-range differences of +25.9 and +28.1 points, so the positive point estimate is not generated solely by near-zero ranges. As a construct sensitivity analysis, the uncalibrated sum of DOVER's technical and aesthetic branches reproduces the earlier +30.3-point contrast, but that sum is neither the official standardized fused score nor a purely technical measure (Table S11).

The reference-model columns give positive point estimates for the substitution, with overlapping intervals across the eligible set. **The supported end-to-end conclusion is that the operation substitution materially changes the selected evidence.**

4.6 Source-level synthesis of the audit

Table 3 brings together the roles the sources document and the results of the implementations evaluated here, organized by work and component.

Three patterns emerge: attribute borrowing, role reassignment, and validation at the summary rather than the component level.

Role reassignment is the one the table makes hardest to miss: *the same computation carries different claims*. Laplacian variance appears twice because two works use it, and the roles are not interchangeable: one applies it as a threshold that discards blurred frames, the other as the dominant weighted term of a keyframe score. The same

Table 3 The audit read by source work. Each band is one source, its role for the component, what this bench measured, and what the source itself reports as evidence; each row is one component. *Dk* = darkening, *Mo* = motion blur, *Bl* = Gaussian blur, *H2* = H.264, *Rs* = rescaling, *St* = desaturation, *Ns* = white noise, *Jp* = JPEG: accuracy (per cent) on the 800 constructed pairs, green where the operation prefers the original and red where it prefers the degraded copy; **50 is the chance level**. **The colour is a background and the value is printed in the cell**, so the table is readable without distinguishing red from green. ***Mean* is over all eight types for every row**, so the column compares like with like; * marks a degradation acting directly on the quantity the operation measures. *Agr*: chance-corrected agreement with LIQE on the operational clusters. *Kp*: percentage of the 42 representatives also chosen by the deployed operation, chance floor 47. The two module rows carry their own sample, stated in the band, and the member shown beside a module is measured on that same sample. † marks the same statistic used by two works, each row on the sample its band states; ‡ marks one colourfulness computation shown in two bands by provenance.

	degradation type										
Source work / component	**Dk**	**Mo**	**Bl**	**H2**	**Rs**	**St**	**Ns**	**Jp**	**Mean**	**Agr**	**Kp**
Ke et al. [32] four features of a naive Bayes photo-quality classifier; *here: each feature alone, as a score;* **human ratings; per feature and combined**											
Edge distribution	70	*62**	*62**	11	74	97	100	31	**63.4**	−.258	**40.5**
Hue count	20	13	10	77	11	4	100	59	**36.8**	−.213	**26.2**
Contrast	*100**	100	100	98	100	30	10	60	**74.8**	+.371	54.8
Colourfulness‡	100	100	100	100	100	*100**	3	79	**85.2**	+.281	52.4
Srinivas et al. [53] averages five attributes into a per-frame quality score; *here: its mean rule, on our attribute vector;* summary-level evaluation											
The mean rule, our vector	100	100	100	100	100	98	29	69	**87.0**	+.416	52.4
Qaroush et al. [47] four terms across three objectives; *here: the module as published, then each term;* summary-level evaluation											
Visual-attention module	100	100	100	100	100	100	0	20	**77.5**	n/a	n/a
Gradient magnitude	100	100	100	100	100	23	0	5	**66.0**	n/a	n/a
Saliency	62	29	20	94	21	18	13	10	**33.4**	n/a	n/a
Contrast term	0	0	0	3	0	68	81	55	**25.9**	n/a	n/a
Entropy	100	77	83	10	77	97	0	66	**63.8**	n/a	n/a
Gradient orientation	100	80	26	95	9	21	0	100	**53.9**	n/a	n/a
Korolkov [33] $.7\,z(\text{sharp}) + .3\,z(\text{bright})$ as a keyframe score, **on 11 of the 20 segments**; *here: the rule, our normalization population;* **no isolated test of the rule**											
The published rule	100	100	100	100	100	0	0	49	**68.6**	n/a	n/a
Sharpness alone, same 11†	100	100	100	91	100	47	0	49	**73.4**	n/a	n/a
Cakmak et al. [8] a threshold discarding blurred frames, not a score; *here: the statistic as a scorer, not the filter;* summary-level evaluation, with ablation											
Laplacian variance†	100	*100**	*100**	95	100	50	0	66	**76.4**	+.551	59.5
Jeong et al. [29] a badness term in a cost multiplicative in cluster distance; *here: the term, read increasing;* summary-level evaluation											
Contrast variance	*100**	100	100	100	100	22	25	90	**79.6**	+.101	64.3
Yang et al. [64] one texture term of several; *here: the term alone;* summary-level evaluation, with ablation											
GLCM contrast	100	100	100	100	100	67	0	72	**79.9**	n/a	n/a
Huang et al. [24] one aesthetic attribute of several; *here: the attribute alone;* summary-level evaluation											
Saturation	13	100	100	15	100	*100**	1	90	**64.9**	−.034	**45.2**
Gygli et al. [20] three aesthetic attributes averaged; *here: the raw mean;* summary-level evaluation, with ablation											
Averaged aesthetics	100	100	100	100	100	100	3	67	**83.8**	n/a	n/a
Hasler and Süsstrunk [21] **the one autonomous index**, and it indexes colourfulness; *here: the index as published;* **human ratings of the named property**											
Colourfulness‡	100	100	100	100	100	*100**	3	79	**85.2**	+.281	52.4

statistic appears in both bands, each row on the sample its band names, and what was measured is the statistic as a scorer. It answers blur on every pair and inverts the ordering under noise. **That characterizes the statistic, not the published filter**: a scorer can order a pair correctly while a threshold keeps or drops both members, and this bench did not run the threshold.

Aggregate masking is visible in the two module rows, whose means sit inside a band of red cells: an aggregate can pass while the types on which a quality score would reject material fail. Section 4.3 gives those figures with their intervals, computed on the types not aligned with each module's own construction and therefore lower than the all-eight means in the table.

Attribute borrowing and the level of validation are what the clause closing each band records, read from the source itself. Of the ten works whose full text is in our archive, two test a measure against human judgment of the property it names. Ke et al. [32] takes community ratings of 60,000 photographs as ground truth and reports per-feature precision–recall alongside the combined classifier, and Hasler and Süsstrunk [21] fits its *colourfulness index* to a psychophysical scaling experiment. **Neither validates a single attribute as a stand-alone score of technical fidelity**: the per-feature curves separate professional photographs from snapshots, which is not the label a selection stage needs. The remaining seven report summary-level evaluation rather than isolated validation of the visual component, and Korolkov [33] fixes its weights without testing them; Table S26 of the supplement gives, source by source, the excerpt that fixes each reading. **Summary-level evaluation does not isolate the visual component's validity as a technical-quality measure.**

Two of the four operations taken from the diversity-and-coverage work serve coverage and diversity there, not quality: the *here:* clause on each band is what separates the reading tested from the job the source gives the operation.

5 Discussion and limitations

Taken together, the results connect provenance to behavior. Operations that agree with one another barely above chance move from heterogeneous source-defined roles into a common technical-quality interpretation; that interpretation fails under controlled degradation; a count-dominance mechanism explains the deployed case, which sits at the median of its own class; and changing the operation changes which representatives the pipeline selects. **That disagreement reaches the ceiling the candidate sets allow**: across the fourteen operations every candidate in every cluster is selected by at least one of them, so the choice of visual-selection operation spans the available decision space rather than perturbing borderline cases. The rest of this section develops those four links.

The first link is documented in Section 3.1 and the second in Section 4.1. For the third, **the measurements establish count dominance** across the evaluated configurations; on the 800 constructed pairs the signs of the score and count differences agree in 797. The count increases observed under noise and JPEG accompany the score inversions, linking the algebraic decomposition to the measured behavior.

Across the other fifteen operations **failures under white noise are frequent**, although their patterns and exceptions do not establish a common mechanism. Figure S2 of the supplement orders the eight degradations by what they do to local high-frequency energy, measured on the same frames as the median ratio to the original. The seven that reduce that energy are answered correctly, from darkening at 84.3 down to desaturation at 61.8. **White noise is the only one that raises it**, by a factor of 6.9, and it is the one the class answers at 19.1 per cent, with a median of 3.0 and thirteen of the fifteen operations below chance. JPEG, which one might expect to add structure through block edges, in fact reduces the measured energy at a ratio of 0.95, and is answered at 62.9. *Structure*, *descriptor count* and *high-frequency energy* are three different quantities, and only the first two were tied together by measurement. **These differences indicate recurring degradation-specific failures** rather than a single mechanism shared by all the evaluated operations.

Validity does not travel automatically when a visual statistic changes task, decision role or aggregation. Attribute borrowing changes the task in which a measure is interpreted; role reassignment changes how it enters the selection; and summary-level evaluation never isolates whether the component measures technical quality. Section 4.6 and Table 3 carry the cases; the controlled and cross-database results are what show that each of those transitions needs its own validation.

For the technical-quality requirement evaluated here, the eligible no-reference models are better-supported replacement candidates than normalizing the deployed operation by segment length or inserting a newer retrieval descriptor. LIQE, QualiCLIP and CLIP-IQA+ meet the common rule and run substantially faster than the deployed path. The complete operation changes 22 decisions, and holding the scorer fixed while changing only the sampled frames and pooling already changes 12; an exact McNemar test separates the two ($p = .006$), and so does the block sign-flip that treats the 9 topics rather than the 42 clusters as the unit ($p = .008$, and $p = .016$ after dropping the topic that contributes most), so both scorer identity and an operation's own input choices carry part of the effect.

5.1 Limitations

Every result above is a statement about one operation, in one reading, under one battery, and none is a verdict on a published system.

Object and transfer. What the bench measured is named beside each row: a term isolated from a joint objective, a module reproduced as published, or an adaptation of ours. A term that fails as a quality score has not thereby been shown to fail at the job its source gives it, and count dominance is established for the deployed aggregation in the configurations evaluated, not for every vocabulary, normalization or extractor.

Application. The 42 clusters are nested in only 19 topics, and both DOVER-technical intervals cross zero: the positive raw, range and rank point estimates, including after removing small-range clusters, do not confirm a population-level output effect. Cluster membership means the pipeline treats segments as redundant alternatives, not that they carry identical completeness, narrative utility, audio quality or semantic relevance, so technical quality remains a tie-breaker rather than an objective

for the summary as a whole. **No human judged these segments**, so what the substitution measures is how far the choice of operation moves the selection, not which selection is preferred.

References and coverage. KADID and the broadcast construction test responses to degradation, not preference between distinct recordings. Q-Align's 98.2% constructed-pair accuracy and the 97.6% of the exact DOVER technical output validate their response to those constructed changes, not their use as human ground truth or as compositional preference models; the official standardized fusion and DOVER++ remain outside the analysis, and DOVER does not isolate temporal modeling because architecture and training differ along with it. The screening rule admits LIQE, QualiCLIP and CLIP-IQA+ without ordering them, so the LIQE substitution demonstrates propagation and not superiority. ARNIQA and TReS are reported on completed axes only and do not enter the complete-case claim, MANIQA is absent, and later models may change the eligible set.

Procedure and reproduction. The screening rule is operational and was not preregistered, and the historical inference environment was not preserved as a fully pinned lockfile, so the archive reproduces the reported statistics without claiming bit-identical regeneration of every neural score. The diagnostic controls are descriptive: the KonIQ–CLIVE–KADID contrasts change content, capture, score distributions and model exposure together, and conditioning on keypoint density may remove legitimate quality signal. Tables S15 and S16 retain those diagnostics; the main conclusion rests on the exact factorization, the dependency-aware external measurements, the constructed pairs, the matched-input control and the controlled operation substitution.

6 Conclusion

This study audited the technical-quality behavior of visual-selection operations drawn from the summarization literature, separating the role each source defines for an operation from the component, module or adaptation evaluated here. Across 800 controlled pairs, **every one of the fifteen operations with a declared direction ranks the degraded copy above the original**, on at least one of eight degradation types, and thirteen do so under white noise. None meets both external requirements, reaching .298 in sensitivity against .50 and .139 in generalization against .65; neither do the nineteen evaluated combinations, of which one clears sensitivity and then fails transfer. Three of ten no-reference models meet both. Where a source specifies how its own terms combine, that rule was implemented as written: of the two complete modules, one scores measurably below its own best component and the other by a margin this sample does not resolve.

For the deployed BoVW+TF-IDF operation an exact factorization separates a descriptor count from a mean inverse-document-frequency weight. **The count dominates in the deployed regime**, where it carries the sign of the score difference, while in the image databases the nineteen convolutional extractors show no comparable grip, the relation survives replacing SIFT with 19 convolutional extractors, and

over the 800 pairs the sign of the score difference follows the sign of the count difference in 797. That explains the inversions under noise and JPEG for one operation; across the other fifteen the outcome recurs as a pattern rather than a demonstrated common cause.

Within the fixed multi-video pipeline, fourteen operations produce eleven distinct representative selections over the same 42 clusters, and replacing the deployed operation with the LIQE-based one **changes 22 of 42 representatives**. These results make **the choice of visual-selection operation an explicit design decision**, to be documented and evaluated against the requirement it is meant to serve. A blind pairwise judgment of real candidates, asking separately about technical quality and about the information each preserves, would decide whether the changed selections yield preferred summaries.

Statements and Declarations

Competing interests

The authors declare no competing financial or non-financial interests.

Data availability

KADID-10k, KonIQ-10k and LIVE In the Wild are public. Supplementary Tables S1–S31, Supplementary Figures S1–S4 and a reproducibility archive accompany the submission; scripts, derived scores, clusters, annotations, seeds and latency results are released at a tagged version of the repository named under Code availability, together with a Git bundle that makes the registered-prediction commit independently verifiable. The HSMVideoSumm news videos remain subject to their original redistribution terms, and only permitted derivatives are archived.

Code availability

Analysis scripts and machine-readable results accompany the submission and are released at https://github.com/Prism411/evaluality-mtap-submission.

Author contributions

J. L. S. G. designed and implemented the evaluation protocol, ran the experiments and wrote the manuscript. R. G. supervised the research, defined the problem framing within the multi-video summarization pipeline and revised the manuscript. Both authors read and approved the final version.

Ethics approval, consent to participate and consent for publication

Not applicable. The study involved no recruitment, intervention or new data collection from human participants, and analyzes existing broadcast material and public image-quality databases.

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